

Bora Atakan

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EDUCATION

University of Waterloo

Waterloo, Ontario, Canada | 2025 – 2030

Bachelor of Computer Science (Honours) with Co-op, GPA: 3.7/4.0

- Courses (Advanced Level): Designing Functional Programs, Calculus 2, Linear Algebra 1

SKILLS

Languages: Python, Racket/Lisp, C, C++, Ruby

Technical Skills: PyTorch, MuJoCo, ROS, W&B, Whisper, Robotics, Welding, Fabrication, SolidWorks, Arduino

EXPERIENCE

University of Waterloo – Robotics Research Intern

June 2026 - Present

- Leading 8-person research team developing end-to-end visuomotor policies. Mentored by [Generalist](#)'s Research Scientist.
- Built and validated the full robot stack: MuJoCo sim environment, data collection, PyTorch training infra, closed-loop inference, dashboards and benchmarking. Got a policy to 99.8% success at 2000+ episodes during an in-person demonstration.
- Built a data pipeline that records synchronized RGB observations, robot poses, and actions at 100 Hz. Collected 13 hours of good quality pick-and-place data.
- Developed and evaluated behavior-cloning policies: DINOv3 backbone with Transformer and CNN/MLP architectures, image and action context window, and action-chunk prediction for closed-loop control.
- Implemented real-time, asynchronous inference that simultaneously runs physics, policy, 100 HZ controls, and logging. Optimized to run on commodity hardware.
- Implemented distributed training infra on A100 + H100 on clusters of Digital Research Alliance of Canada using SLURM, W&B, EMA checkpoints, and resumable training. Accelerated training throughput 9× by caching vision encoder embeddings.
- Built deterministic checkpoint benchmarking and Rerun visualization tools to compare policies on seeded randomized episodes and inspect camera observations, 3D trajectories, predicted action horizons, executed commands, and success rates.

Thresh Power – Software Development Intern

Summer 2024

- Worked on search and information retrieval. Gathered requirements and planned project under mentorship, then independently shipped backend and frontend features used by paying customers.
- Shipped full-text search feature using Elasticsearch and Python, indexing thousands of PDF documents. This feature was leveraged by the company to secure seed round funding and attract new customers.
- Consolidated search under a single API and hand-tuned the scores to appropriately weigh different search dimensions (file name, summary, content, etc.). Implemented search result highlight using React.

PROJECTS

Minecraft RL Environment

- Built a 3D video game environment, simplified Minecraft clone, with seed-based deterministic environment generation, physics, and interactive objects.
- Implemented headless mode for screenshot based regression testing.
- Built a replay feature and RL data collection mode to build a dataset and fine-tune ViT backbones for real-time action chunking control models.
- Built an admin control UI in React and FastAPI.

CUDA Kernel Dev Environment

- Built an RL environment for CUDA kernel development for fused machine learning operations (LayerNorm, Residual, SiLU). Built a judge (correctness + performance harness) on A10 GPUs using Modal, nvcc (identify register spills), and dataset of correct input-output. Got 20% success rate on Claude Haiku 4.5 with 5 tries for each attempt over 100 attempts.

Speech-to-phonemes

- Created 183 GB Hugging Face dataset for phoneme detection. Created mixed dataset dataloader and ran training jobs on A10 and A100 GPUs on Lambda. Improved accuracy metrics in end-to-end evals.

EXTRACURRICULAR

RedFox Robotics

2022 – Present

- Co-founded RedFox Robotics and co-led a team of 26. Raised over \$10,000 for workshop; trained 23 new members; created simulations for faster iteration on robot controls; placed 11th at regional qualification, and advanced to the playoffs as Alliance 7's captain.